Project:	Beneath Every Tide (BET)
Team No.:	Team #2
Class:	CSE 3310; Fall 2026
Module:	System Requirements Analysis (SRA)
Deliverable:	SRA Document

Version: [1.0] **Date: [10/29/2026]**

Contributors:

Hiya Kavalanekar
Prasil Niraula
Alyssa Aguilar
Jayneshkumar Patel

Revision History

Version number	***Date***	***Originator***	***Reason for change***	***High-level description of changes***
1.0	10/29/2026	Jayneshkumar Patel	Initial draft	GPS has been removed, Scoring UML has changes

TABLE OF CONTENTS

1. INTRODUCTION AND PROJECT OVERVIEW 3

2. OBJECTIVES 4

2.1 BUSINESS Objectives 4

2.2 SYSTEM Objectives 5

3. PROJECT CONTEXT DIAGRAM 6

4. SYSTEMS REQUIREMENTS 7

4.1 “Scoring” Requirements 7

4.2 “Win/Lose” Requirements 11

4.3 “Setup” Requirements 15

4.4 “AI” Requirements 21

4.5 “Graphics” Requirements 31

5. SOFTWARE PROCESSES AND INFRASTRUCTURE 44

5.1 Hardware and Infrastructure 44

5.2 UML Diagrams 44

5.3 Conceptual Data Model 49

5.4 Screen Shots 49

5.5 Test Plan 49

6. ASSUMPTIONS AND CONSTRAINTS 50

6.1 ASSUMPTIONS 50

6.2 CONSTRAINTS 51

6.3 Out-of-Scope Material 51

7. DELIVERY AND SCHEDULE 52

8. STAKEHOLDER APPROVAL FORM 53

APPENDIX: 54

1. Introduction and Project Overview

Beneath Every Tide (BET) is a single-player, touch-controlled fishing game for Android. The player casts bait, hooks fish, and reels them in across three fixed levels, each with its own map. Every confirmed catch earns money directly. There is no shop and no conversion step. Each level has a cash target: $10,000 for Level 1, $15,000 for Level 2, and $25,000 for Level 3.

The player has a ten-minute active-play timer per level. A level ends immediately in a win when its target is met. It ends in a loss if the timer reaches zero first. Catching trash costs $100, and earnings can never drop below $0. Catching several fish of the same type in a row earns cumulative combo bonuses. Winning all three levels wins the game, with at least $50,000 banked. Failing a level only resets that level's earnings, so money from completed levels is kept.

The game has Settings for brightness, volume, difficulty (Easy/Medium/Hard), and touch controls. This SRA defines the requirements for five components: Scoring, Win/Lose, Setup/Settings/Game Controller, AI, and Graphics. Each has a single owner, so the rules for money, outcomes, behaviour, and presentation never overlap.

2. Objectives

2.1 BUSINESS OBJECTIVES

The following is a list of business objectives:

Objective 1: Graphics: The game shall present the fishing experience clearly on an Android touch screen. It shows one fixed map for each of the three levels and visually distinct Basic, Royal, and Elite fish and trash. It displays a live HUD with the current level, level target, current earnings, banked total, and time remaining. It also provides cast, hook, and reel controls with hook prompts, fishing animations, payout and combo notifications, the settings screens, and the Level Won, Level Lost (Time's Up), and Game Won screens. Graphics only displays state supplied by other components and never makes gameplay decisions.

Objective 2: Setup (Settings and GameController): The game shall let the player customize and control the session. Players can adjust brightness, music and sound-effect volume, difficulty (Easy, Medium, Hard), control sensitivity, and button size. Exit saves a change, Cancel discards it, and accepted settings persist across relaunches. The game controller shall run the three fixed levels and maps and manage the ten-minute active-play timer. It shall pause gameplay while Settings is open and move the player through next level, retry, and play again.

Objective 3: Win or Lose: The game shall decide each level's outcome from the level's cash target. A level is won the moment earnings reach $10,000, $15,000, or $25,000 for Levels 1, 2, and 3, without waiting for the timer. A level is lost only if time expires before the target is met. Winning all three levels wins the game with at least $50,000 banked. Each attempt produces exactly one result.

Objective 4: Scoring: The game shall reward the player with money directly for each confirmed catch, with no shop or conversion step. Basic fish pay $100, Royal $150, and Elite $250. Consecutive same-type catches earn cumulative combo totals (Basic $250 then $400, Royal $400 then $550, Elite $600 then $850), and only the difference is credited each time. Collected trash costs $100, never takes earnings below $0, and resets the combo. Current-level earnings are kept separate from banked earnings, so a failed level never costs the player money from completed levels. Scoring is the only component that changes money, and each event is counted once.

Objective 5: AI: The game shall provide fishing gameplay through fish and trash behavior. When the player casts, AI detects and classifies a Basic, Royal, or Elite fish or a piece of trash. A fish may approach and bite, and the player must hook it within a difficulty-based time window. During the struggle the fish has a type-based chance to escape. AI reports each completed fish catch or trash collection once, without calculating any money. Difficulty changes fish speed, hook timing, and spawn frequency, and fishing resumes automatically after each encounter.

2.2 SYSTEM OBJECTIVES

The following is a list of system objectives:

Objective 1: The system will be an Android-based mobile game with a touch interface, run on the Android emulator required by CSE 3310.

Objective 2: The system will be built with Unity, a provisional choice (A18). The functional requirements do not depend on the engine.

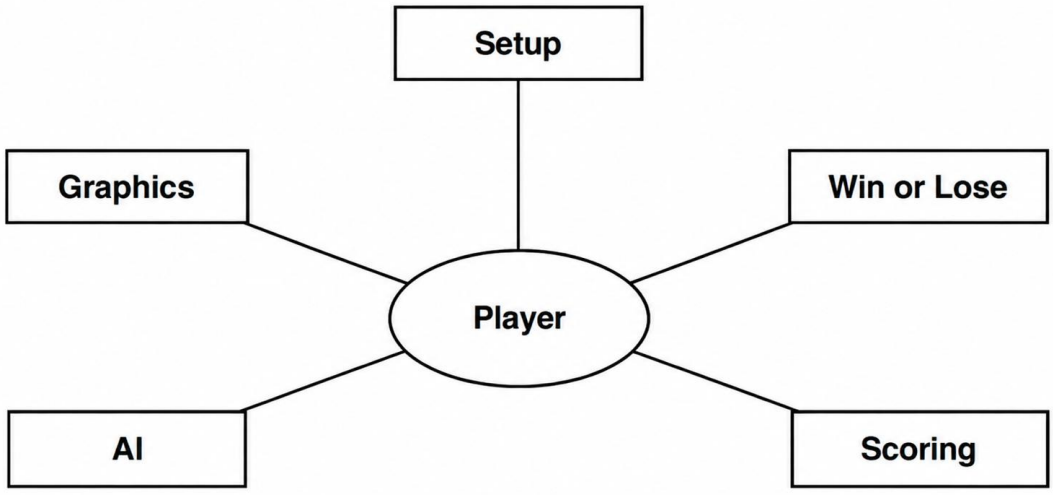
Objective 3: The system will be organized into six components with defined ownership: Game Controller, Settings, AI, Scoring, Win/Lose, and Graphics.

Objective 4: Components will communicate through typed events: Catch Confirmed, Trash Collected, Score Changed, Level Won, Level Lost, Game Won, Time Expired, Fish Bitten, and Escape Occurred. Each encounter and attempt carries a unique ID so every event is processed exactly once.

Objective 5: Game values will be configurable parameters rather than hard-coded: timer length, fish and trash probabilities, difficulty multipliers, hook windows, and spawn intervals.

Objective 6: Accepted Settings values will be stored on the device and reloaded at relaunch. Game progress is kept only for the current runtime session (A10).

3. Project Context Diagram



4. Systems Requirements

4.1 "SCORING" REQUIREMENTS

Requirement Title: (*required)	Award dollars for confirmed fish catches
Sequence No: (*required)	001
Short description: (*required)	The game awards points to the player when a fish is successfully caught based on which fish is caught.
Detailed Description: (*required)	When a player's rod successfully catches an object, the system identifies whether the caught object is a fish or trash. If the object is identified as fish, the type of fish is checked. According to the type of fish, the associated points are added to the total earned points by the player and is displayed. If the player has caught fish consecutively, the system evaluates whether the player qualifies for a combo bonus.
Pre-Conditions: (optional)	1. The game has started, and the player has started fishing. 2. The player has successfully caught an object. 3. The system can identify the object as fish.
Post Conditions: (optional)	1. The points earned for the caught fish are added to player's current score 2. The system can evaluate the combo bonus addition.
Other attributes: (optional)	- The points awarded depend on the type of fish caught

Requirement Title: (*required)	Apply Penalty when TRASH is caught
Sequence No: (*required)	002
Short description: (*required)	The system deducts points from the player's score when the player catches trash.
Detailed Description: (*required)	When a player's rod catches an object, the system identifies whether the caught object is a fish or trash. If the object is identified as trash, the system shall subtract the applicable penalty points from the player's current score and reset the consecutive fish-catching streak. The system shall then update the score displayed and the game continues.
Pre-Conditions: (optional)	1. The game has started, and the player has started fishing. 2. The player has successfully caught an object. 3. The system can identify the object as fish.
Post Conditions: (optional)	- None
Other attributes: (optional)	The penalty amount is applied according to the game's scoring rules.

Requirement Title: (*required)	Scoring
Sequence No: (*required)	003
Short description: (*required)	Application of Combo Bonus
Detailed Description: (*required)	If a player catches 3 fishes consecutively without catching even a single trash then a combo bonus is applied. Different types of bonuses are applied based on which types of fish are caught. The patterns include: Basic + Basic + Basic = $100 Basic + Basic + Royal = $130 Basic + Basic + Elite = $160 Basic + Royal + Royal = $200 Basic + Elite + Elite = $300 Elite + Royal + Royal = $410 Elite + Elite + Royal = $550 Elite + Elite + Elite = $777
Pre-Conditions: (optional)	1. The game is in an active fishing round. 2. The player has successfully caught three fish consecutively without catching any trash.
Post Conditions: (optional)	1. The system applies the combo bonus corresponding to the combination of the three fish caught. 2. The bonus is added to the player's score.
Other attributes: (optional)	- A combo bonus shall be awarded only when three fish are caught consecutively without catching trash. Catching trash shall reset the consecutive fish-catching streak.

Requirement Title: (*required)	Scoring
Sequence No: (*required)	004
Short description: (*required)	Check Goal & Remaining Time
Detailed Description: (*required)	After every fish caught by the player the system should check where the current score stands as well as the remaining time. If the current score is greater than or equal to the goal then the level is cleared and the player moves on to the next level. However, if the current score is less than the goal required **AND** the time has runover then the player fails that level and needs to replay in order to move forward.
Pre-Conditions: (optional)	1. The player is participating in an active fishing round. 2. The system has access to the player's current score, the goal score, and the remaining time.
Post Conditions: (optional)	- The system proceeds according to the outcome of the goal and time checks.
Other attributes: (optional)	- The system accesses both the timer and the goal score after the fish is caught.

4.2 "WIN/LOSE" REQUIREMENTS

Requirement Title: (*required)	Winner or Loser
Sequence No: (*required)	001
Short description: (*required)	Determining Level Win
Detailed Description: (*required)	The game shall receive the calculated score from the scoring component and identify the level completed and its required score. If the calculated score is equal to or more than the required score for that level then it's a win and the player has successfully completed that level and can move on to the next level or win the entire game if the level completed was the final level. The message saying "Winner" shall be displayed on the screen along with the required and achieved score.
Pre-Conditions: (optional)	1. The player has completed the current level and the score for the round has been calculated. 2. The system is able to access the calculated score from "Scoring"
Post Conditions: (optional)	- The system is able to load graphics and AI for displaying the message "Level Won!" and to load the next level.
Other attributes: (optional)	- The player only moves forward if the required score is met either equal to or greater.

Requirement Title: (*required)	Winner or Loser
Sequence No: (*required)	002
Short description: (*required)	Determining Level Loss
Detailed Description: (*required)	The game shall receive the calculated score from the scoring component and identify the level completed and its required score. If the calculated score is less than the required score for that level then the player has been defeated and the player has to retry the same level in order to move forward. The message saying “Level Lost!” shall be displayed on the screen along with the required and achieved score. .
Pre-Conditions: (optional)	1. The player has completed the current level and the score for the round has been calculated. 2. The system is able to access the calculated score from “Scoring”
Post Conditions: (optional)	- .The system is able to load graphics and AI for displaying the message “Level Lost!” and to load back the same level
Other attributes: (optional)	- Retrying should load the same level again rather than advancing to the next level.

Requirement Title: (*required)	Determine final victory after Level 3
Sequence No: (*required)	003
Short description: (*required)	Finish the game only after all three level goals have been met.
Detailed Description: (*required)	When a valid Level Won is confirmed for Level 3 and Levels 1 and 2 have already been won, Game Controller shall first authorize Scoring to bank Level 3 earnings. After banking completes, Win/Lose Manager shall emit GameWon(totalBankedEarnings, completedLevels=3). Because each required level amount contributes to the goal, total banked money must be at least $50,000. If an individual level overshoots its target, the higher actual balance may be shown; the game does not require exactly $50,000.
Pre-Conditions: (optional)	Level 3 has been won following valid recorded wins for Levels 1 and 2.
Post Conditions: (optional)	Final Game Won result is emitted once and no fourth level is requested.
Other attributes: (optional)	- None

Requirement Title: (*required)	Provide consistent result details to Graphics
Sequence No: (*required)	004
Short description: (*required)	Give values and result type, without drawing screens or launching AI.
Detailed Description: (*required)	For Level Won, Level Lost, or Game Won, Win/Lose Manager will publish a typed result containing levelId, target, achieved Level Earnings, total Earnings/total, and result Type. It must NOT render the “Level Won!”, “Level Lost!”, “Time’s Up!” or “Game Won!” screens, and should NOT initialize AI, animations, timers, or maps; these responsibilities belong to Graphics and Game Controller. Game Controller shall tell Graphics when to present the result.
Pre-Conditions: (optional)	A win/loss/final result has been determined.
Post Conditions: (optional)	A complete result object exists for the presentation layer and next-level workflow.
Other attributes: (optional)	- None

4.3 “SETUP” REQUIREMENTS

Requirement Title: (*required)	Setup
Sequence No: (*required)	001
Short description: (*required)	Open Settings
Detailed Description: (*required)	When the user clicks the settings menu, it will pause the game and open the settings menu to display all settings options. The user can press: • The gear icon within the game
Pre-Conditions: (optional)	The game must be paused before the user can open settings
Post Conditions: (optional)	None
Other attributes: (optional)	None

Requirement Title: (*required)	Setup
Sequence No: (*required)	002
Short description: (*required)	Settings Menu
Detailed Description: (*required)	The user can choose the setting they wish to adjust. The user can press: • Volume (change the volume) • Brightness (change the brightness) • Difficulty (change the difficulty) • Controls (change the controls) • Exit
Pre-Conditions: (optional)	- The user must have opened the settings menu.
Post Conditions: (optional)	- None
Other attributes: (optional)	- None

Requirement Title: (*required)	Setup
Sequence No: (*required)	003
Short description: (*required)	Control brightness
Detailed Description: (*required)	The user can adjust the brightness of the game using a slider. The user can (press): • Use the slider to adjust brightness • Cancel (discard changes) • Reset to default brightness • Adjust controls • Exit (go back to settings menu)
Pre-Conditions: (optional)	• The user must have opened the settings menu • The user must have clicked the Brightness button
Post Conditions: (optional)	The adjustments made by the user are saved automatically on exit
Other attributes: (optional)	-None

Requirement Title: (*required)	Setup
Sequence No: (*required)	004
Short description: (*required)	Change volume
Detailed Description: (*required)	The user can adjust the volume of the music and sound effects in the game with a slider. The user can (press): • Adjust the music volume slider • Adjust the sound effect volume slider • Cancel (discard changes) • Reset to default volume • Exit (go back to settings menu)
Pre-Conditions: (optional)	- The user must have opened the settings menu - The user must have clicked the Volume button
Post Conditions: (optional)	The adjustments made by the user are saved automatically on exit
Other attributes: (optional)	- None

Requirement Title: (*required)	Setup
Sequence No: (*required)	005
Short description: (*required)	Change difficulty
Detailed Description: (*required)	The user can choose the difficulty of the game The user can press/choose: • Easy difficulty • Medium difficulty • Hard difficulty • Cancel (discard changes) • Reset to medium difficulty • Exit (go back to settings menu)
Pre-Conditions: (optional)	• The user must have opened the settings menu • The user must have clicked the Difficulty button
Post Conditions: (optional)	The adjustments made by the user are saved automatically on exit
Other attributes: (optional)	- None

Requirement Title: (*required)	Setup
Sequence No: (*required)	006
Short description: (*required)	Change controls
Detailed Description: (*required)	The user can adjust the sensitivity or size for the controls of the game using a slider. The user can (press): • Easy difficulty • Adjust the sensitivity slider • Adjust the button size slider • Cancel (discard changes) • Reset to default controls • Exit (go back to settings menu)
Pre-Conditions: (optional)	• The user must have opened the settings menu • The user must have clicked the Controls button
Post Conditions: (optional)	The adjustments made by the user are saved automatically on exit
Other attributes: (optional)	- None

4.4 “AI” REQUIREMENTS

Requirement Title: (*required)	Casting Bait and Object Detection
Sequence No: (*required)	001
Short description: (*required)	Identify whether the cast bait encounters a fish or trash.
Detailed Description: (*required)	1. The system shall begin the detection process when the player casts bait into the fishing area. 2. The AI shall transition from Idle to Determine Object and inspect the object interacting with the bait. 3. If the detected object is a fish within the applicable detection range, the system shall proceed to Fish Detects Bait. 4. If the object is trash, the system shall route the interaction to Trash Hooked rather than initiating fish behavior. 5. If no valid object is detected, the system shall remain ready to evaluate subsequent bait interactions.
Pre-Conditions: (optional)	1. The game session is active, and bait casting is available. 2. The fish AI is in Idle before the interaction begins.
Post Conditions: (optional)	1. The encountered object is classified and routed to the appropriate behavior. 2. The system does not trigger fish-bite behavior for trash.
Other attributes: (optional)	- Exact detection distances and collision criteria will be defined during implementation.

Requirement Title: (*required)	Fish AI - Bait Detection
Sequence No: (*required)	002
Short description: (*required)	Evaluate whether a nearby fish is attracted to the bait.
Detailed Description: (*required)	1. When a fish is identified near the cast bait, the system shall enter the Fish Detects Bait state. 2. The AI shall evaluate whether the fish responds to the bait according to the attraction logic implemented for the game. 3. If attracted, the fish shall change state to Fish Approach and begin moving toward the bait. 4. If the fish ignores the bait, it shall leave the interaction and return to Idle. 5. The system shall communicate the relevant detection response to the player without declaring a catch prematurely.
Pre-Conditions: (optional)	1. Bait is present in the fishing area. 2. A fish has been detected within the configured range.
Post Conditions: (optional)	1. An attracted fish enters Fish Approach. 2. A fish that does not respond returns to Idle.
Other attributes: (optional)	- Attraction conditions and any probability values require team approval and tuning.

Requirement Title: (*required)	Fish Approach and Bite
Sequence No: (*required)	003
Short description: (*required)	Move an attracted fish toward the bait and initiate a bite.
Detailed Description: (*required)	1. After successful bait detection, the AI shall place the fish in Fish Approach. 2. The fish shall move toward the bait while the bait remains available in the fishing area. 3. When the fish reaches the bait, the AI shall transition to Fish Bites and initiate the bite interaction. 4. The game shall notify the player that the fish has bitten and that a hook-setting action is needed. 5. If the bait is removed before the fish reaches it, the AI shall cancel the approach and return to Idle.
Pre-Conditions: (optional)	1. A fish has reacted positively to the bait. 2. The bait is still present when the approach begins.
Post Conditions: (optional)	1. A successful approach enters Fish Bites and gives the player a hook prompt. 2. An interrupted approach ends without a bite.
Other attributes: (optional)	- Fish movement speed and the distance required to count as reaching bait are design parameters.

Requirement Title: (*required)	Hook Timing
Sequence No: (*required)	004
Short description: (*required)	Determine whether the player sets the hook successfully.
Detailed Description: (*required)	1. Upon entering Fish Bites, the system shall display or otherwise provide a clear prompt to set the hook. 2. The game shall evaluate the player's hook-setting input against the permitted response window. 3. If the input is valid and timely, the fish shall become hooked, and the AI shall transition to Struggle. 4. If the player misses the valid window or fails to set the hook, the AI shall transition to Escape. 5. The system shall provide feedback indicating whether the hook action succeeded or the fish escaped.
Pre-Conditions: (optional)	1. The fish is in Fish Bites. 2. The player has received the hook prompt.
Post Conditions: (optional)	1. Successful timing begins with Struggle. 2. Unsuccessful timing results in Escape with player feedback.
Other attributes: (optional)	- The hook timing window and exact input method must be finalized by the team

Requirement Title: (*required)	Fish Struggle and Escape
Sequence No: (*required)	005
Short description: (*required)	Simulate resistance after hooking and resolve the catch attempt.
Detailed Description: (*required)	1. Once the hook is set, the AI shall enter Struggle and animate fish resistance during the catch attempt. 2. The fish shall exhibit changing or unpredictable movement to distinguish a hooked fish from an idle fish. 3. The game shall evaluate the player's actions against the conditions required to keep the fish hooked. 4. If the player successfully completes the struggle, the system shall transition the fish to Caught. 5. If the hook is lost or the player fails the struggle, the system shall transition to Escape and notify the player.
Pre-Conditions: (optional)	1. The hook-setting action has succeeded. 2. The fish is in Struggle and remains part of an active catch attempt.
Post Conditions: (optional)	1. The catch attempt ends with either Caught or Escape. 2. Only a Caught result is eligible for fish-point processing.
Other attributes: (optional)	- Detailed controls, resistance levels, and win/fail conditions are to be specified in game design.

Requirement Title: (*required)	Caught Fish and Points
Sequence No: (*required)	006
Short description: (*required)	Record a successful catch and award the corresponding fish points.
Detailed Description: (*required)	1. After the player wins the struggle, the system shall transition the fish from Struggle to Caught. 2. The game shall notify the player of the successful catch and identify the caught fish as the scoring event. 3. The system shall move to Update Score and add the assigned point value for the fish. 4. Each completed catch shall be processed once so that a single fish does not award points repeatedly. 5. After scoring, the game shall make the continue-fishing decision available to the player
Pre-Conditions: (optional)	1. The fish has reached Caught following a successful struggle. 2. A score value is associated with the caught fish.
Post Conditions: (optional)	1. The fish catch is reflected in the updated score. 2. The player can decide whether to cast bait again.
Other attributes: (optional)	- Fish-point values and any relationship to fish types will be defined by the team.

Requirement Title: (*required)	Trash Hooking and Collection
Sequence No: (*required)	007
Short description: (*required)	Handle trash encounters and apply the associated score penalty.
Detailed Description: (*required)	– When Determine Object identifies trash, the system shall transition to Trash Hooked rather than using fish-bite states. – The game shall notify the player that the bait has hooked a trash object. – When the player reels in the trash, the system shall enter Trash Collect and register the collected object. – The system shall transition to Update Score and subtract the configured trash penalty. – After the penalty is processed, the game shall offer the player the option to continue fishing.
Pre-Conditions: (optional)	– The active bait interaction has been classified as trash. – The game supports a trash-reeling interaction.
Post Conditions: (optional)	- The collected trash is registered. - The current score reflects the trash penalty after updating.
Other attributes: (optional)	- The exact penalty and any disposal or inventory mechanics are not yet specified

Requirement Title: (*required)	Resume fishing automatically after resolved interaction
Sequence No: (*required)	008
Short description: (*required)	Return to Idle without a Continue Fishing menu.
Detailed Description: (*required)	After confirmed fish catch, collected trash, an escaped fish, or an empty encounter, AI shall release the bait lock and return to Idle automatically if GameController status is Playing and time remains. It shall never ask the player to choose Continue Fishing after every catch. If the attempt has ended or Settings is open, AI shall remain suspended or canceled as directed instead of starting a new encounter.
Pre-Conditions: (optional)	An encounter has ended and its result was processed or canceled.
Post Conditions: (optional)	Player can cast again immediately during active gameplay, with no continue prompt.
Other attributes: (optional)	- None

Requirement Title: (*required)	Pause, cancel, and prevent late AI results
Sequence No: (*required)	009
Short description: (*required)	Honor Settings pause, level completion, retry, and time expiration.
Detailed Description: (*required)	- On Paused, AI shall freeze fish movement, hook windows, struggle progress, and spawn timers. - On Time Expired, Level Won, Level Lost, or game exit, AI shall terminate current encounter and stop updating the fishing scene. - AI shall not emit new Catch Confirmed or Trash Collected events after controller closes the attempt and shall ignore any old attempt after Retry or Next Level.
Pre-Conditions: (optional)	- Controller emits pause/resume or terminal state transitions.
Post Conditions: (optional)	- No fish continues moving or scores after timeout or level transition, paused struggle resumes without losing time.
Other attributes: (optional)	- None

Requirement Title: (*required)	Apply difficulty to movement, timing, and spawn frequency
Sequence No: (*required)	010
Short description: (*required)	Use selected difficulty without changing score calculations.
Detailed Description: (*required)	- AI shall accept Settings difficulty and apply the current movement multiplier, hook window, and spawn interval. - Proposed Easy = 0.75x / 3s / 2s; Medium = 1.0x / 2s / 3s; Hard = 1.25x / 1s / 4s, where each row means movement multiplier / hook window / spawn attempt interval. - These values shall be configurable, and no AI behavior shall change fish prices, cash targets, or scoring combo formulas.
Pre-Conditions: (optional)	- Settings selected, AI initialized or resuming active play.
Post Conditions: (optional)	- Changing Easy/Medium/Hard changes the intended three AI behaviors while monetary rules remain constant.
Other attributes: (optional)	- None

4.5 “GRAPHICS” REQUIREMENTS

Requirement Title: (*required)	GraphicsManager
Sequence No: (*required)	001
Short description: (*required)	Coordinate all visual components without owning game rules.
Detailed Description: (*required)	- GraphicsManager shall initialize MainDisplay, ScoreDisplay, Renderer, GameObject presentation, Fish/Trash appearance, three fixed Maps, Settings Menu, and Animation resources. - It shall subscribe to read-only gameplay updates from GameController, AI, Scoring, and Win/Lose and request appropriate redraws. - It shall not modify dollars, alter fish movement, or decide win/loss results.
Pre-Conditions: (optional)	- GameController has initialized session and graphics dependencies.
Post Conditions: (optional)	- Graphics components are active and show the provided state.
Other attributes: (optional)	- None

Requirement Title: (*required)	Main Display
Sequence No: (*required)	002
Short description: (*required)	Render the primary fishing scene for the active level.
Detailed Description: (*required)	- MainDisplay shall present the fixed level map, water/fishing scene, bait, player fishing controls, fish/trash objects and current HUD. - It shall refresh as supplied gameplay state changes; it shall not run the timer, interpret catch rules, or control fish movement. - It must remain visible as a frozen background behind the Time's Up! overlay.
Pre-Conditions: (optional)	- GraphicsManager initialized, GameController selected level/map.
Post Conditions: (optional)	- The gameplay scene corresponds to the latest active or frozen state.
Other attributes: (optional)	- None

Requirement Title: (*required)	Score Display
Sequence No: (*required)	003
Short description: (*required)	Always show level target, current earnings, banked earnings, time, and level.
Detailed Description: (*required)	- ScoreDisplay shall show current level number (1/2/3), levelEarnings in $USD, level target ($10,000/$15,000/$25,000), bankedEarnings or totalDisplayed toward $50,000, and time remaining in mm:ss. - When Scoring emits ScoreChanged or GameController emits TimerChanged, it shall refresh these values without calculating or mutating them. - It shall indicate current combo status when provided.
Pre-Conditions: (optional)	- MainDisplay active, read-only score and level/time snapshots exist.
Post Conditions: (optional)	- Current money, target, timer, level, and relevant combo information are readable.
Other attributes: (optional)	- None

Requirement Title: (*required)	Renderer
Sequence No: (*required)	004
Short description: (*required)	Draw current GameObject objects without implementing movement AI.
Detailed Description: (*required)	- Renderer shall draw all visible fish, trash, bait, and other GameObject instances at positions provided by their authoritative game-state owners - It shall update orientation, visibility, and appearance as state changes and stop movement playback when gameplay is paused or frozen. - It shall not decide if fish bites, escapes, or is caught.
Pre-Conditions: (optional)	- Current scene and list of visible object states exist.
Post Conditions: (optional)	- All visible objects appear at correct supplied positions, and hidden/removed objects do not render.
Other attributes: (optional)	- None

Requirement Title: (*required)	GameObject
Sequence No: (*required)	005
Short description: (*required)	Provide shared graphical representation for displayable objects.
Detailed Description: (*required)	- Each Game Object presentation shall expose at least a visual identity, objectId, objectType, position, visibility, and sprite/animation state - Fish and Trash presentations shall reuse these shared rendering attributes. - User identifies the visual object and AI logs the logical catch attempt and shall not be used to calculate money.
Pre-Conditions: (optional)	- Game object was created or registered by graphics.
Post Conditions: (optional)	- Renderer can draw and update fish/trash without embedding game rules.
Other attributes: (optional)	- None

Requirement Title: **(*required)**	Fish
Sequence No: **(*required)**	006
Short description: **(*required)**	Show identifiable visual types and AI-driven fish states.
Detailed Description: **(*required)**	Fish rendering shall distinguish Basic Fish, Royal Fish and Elite Fish and shall display state changes provided by AI: Idle, Fish Approach, Fish Bites, Struggle, Caught, or Escape. Presentation shall display movement and animations but shall not select fish rarity, compute escape chances, or award dollars.
Pre-Conditions: **(optional)**	AI has supplied fishType and current fish state.
Post Conditions: **(optional)**	A fish is visibly identifiable and its visual state follows AI.
Other attributes: **(optional)**	- None

Requirement Title: **(*required)**	Basic Fish
Sequence No: **(*required)**	007
Short description: **(*required)**	Render Basic category consistently.
Detailed Description: **(*required)**	Basic Fish shall use an appearance distinguishable from Royal Fish and Elite Fish, update position from AI, and support bite/struggle/caught/escape animation requests. Its money value is $100.
Pre-Conditions: **(optional)**	An AI fish with fish Type Basic is present.
Post Conditions: **(optional)**	Basic fish appears with its own identity and follows supplied movement state.
Other attributes: **(optional)**	- None

Requirement Title: **(*required)**	Royal Fish
Sequence No: **(*required)**	008
Short description: **(*required)**	Render Royal category consistently.
Detailed Description: **(*required)**	Royal Fish shall use different appearance from Basic Fish and Elite Fish, update position from AI, and support bite/struggle/caught/escape animation requests. Its money value is $150.
Pre-Conditions: **(optional)**	An AI fish with fish Type Royal is present.
Post Conditions: **(optional)**	Royal fish appear with its own identity and follows supplied movement state.
Other attributes: **(optional)**	- None

Requirement Title: **(*required)**	Elite Fish
Sequence No: **(*required)**	009
Short description: **(*required)**	Render Elite category consistently.
Detailed Description: **(*required)**	Elite Fish shall use a different appearance and distinguishable from Basic Fish and Royal Fish, update position from AI, and support bite/struggle/caught/escape animation requests. Its money value is $250.
Pre-Conditions: **(optional)**	An AI fish with fish Type Elite is present.
Post Conditions: **(optional)**	Elite fish appear with its own identity and follows supplied movement state.
Other attributes: **(optional)**	- None

Requirement Title: **(*required)**	**Trash**
Sequence No: **(*required)**	010
Short description: **(*required)**	Render trash distinct from fish and show collection feedback.
Detailed Description: **(*required)**	Trash shall be visually different from all fish types. It shall appear or disappear according to AI object state and play a collected-state cue when AI confirms collection. The $100 penalty and zero loss if the level earning is zero which should be provided only to Scoring. Graphics shall display a -$100 indication only from a committed score event.
Pre-Conditions: **(optional)**	AI has supplied trash object and/or collection state.
Post Conditions: **(optional)**	Trash is visually recognizable and confirmed penalty events are displayed accurately.
Other attributes: **(optional)**	- None

Requirement Title: **(*required)**	Map (Three Fixed Fishing Scenes)
Sequence No: **(*required)**	011
Short description: **(*required)**	Show one predetermined map for each of the three game levels.
Detailed Description: **(*required)**	Map shall render Map 1 for Level 1, Map 2 for Level 2, and Map 3 for Level 3 as commanded by Game Controller. It shall not retrieve device location, choose geolocation or zones. It shall refresh scene/map assets on authorized level transitions while the prior level is inactive.
Pre-Conditions: **(optional)**	GameController has supplied active levelId and mapId.
Post Conditions: **(optional)**	The correct fixed fishing map appears for the level.
Other attributes: **(optional)**	- None

Requirement Title: **(*required)**	SettingsMenu
Sequence No: **(*required)**	012
Short description: **(*required)**	Display Brightness, Volume, Difficulty, Controls, Exit and Cancel interfaces.
Detailed Description: **(*required)**	SettingsMenu shall render the settings main menu and submenus with visible sliders, choices, Reset, Cancel, and Exit controls. It shall forward user interactions to Setup component and display current draft or saved values returned by Setup. The UI shall use the accepted button-size and sensitivity configuration. It should not save preferences directly or determine gameplay difficulty.
Pre-Conditions: **(optional)**	Controller has paused gameplay and Settings view model exists.
Post Conditions: **(optional)**	UI accurately reflects Setup-supplied values and input actions reach Settings.
Other attributes: **(optional)**	- None

Requirement Title: **(*required)**	Animation
Sequence No: **(*required)**	013
Short description: **(*required)**	Play and stop visual gameplay animations on authoritative events.
Detailed Description: **(*required)**	Animation shall support cast, fish approach, bite, struggle, catch, escape, trash collection, and combo feedback. AI controls the underlying behavior/outcome, Animation only plays the requested sequence. Pause shall freeze animations without losing state, timeout or level completion shall stop gameplay animations immediately. The screen may remain visually frozen behind result overlays.
Pre-Conditions: **(optional)**	A corresponding state and animation asset exist.
Post Conditions: **(optional)**	The requested animation shows or stops in sync with controller/AI state.
Other attributes: **(optional)**	- None

Requirement Title: **(*required)**	On-screen fishing controls and hook prompts
Sequence No: **(*required)**	014
Short description: **(*required)**	Render Cast, Hook, Reel controls and immediate fish-bite prompts.
Detailed Description: **(*required)**	Main Display shall show the on-screen fishing controls agreed for the touch interface, including Cast when idle, an obvious Hook prompt when AI emits Fish Bitten, and reeling feedback while a fish is in Struggle. Controls shall send user actions to GameController, not directly to Scoring. When Settings is open or gameplay ended, fishing buttons shall be disabled/hidden.
Pre-Conditions: **(optional)**	Game Controller says Playing and supplies the current fishing interaction state.
Post Conditions: **(optional)**	Fishing input is discoverable and the player receives a timely hook prompt.
Other attributes: **(optional)**	- None

Requirement Title: **(*required)**	Fish payout, combo, and trash notifications
Sequence No: **(*required)**	016
Short description: **(*required)**	Explain money changes without recalculating payouts in the UI.
Detailed Description: **(*required)**	When a committed ScoreChanged arrives, Graphics shall show the fish type or trash event, incremental change (e.g. Basic third catch +$150), current streak (2x/3x), and Total streak award when applicable Basic 3x = $400 Royal 3x = $550 Elite 3x = $850 A confirmed collected-trash event may show -$100 or the actual clamped reduction at $0 (which can be $0). All displayed monetary values shall come from Scoring. The indicator shall not pause the session.
Pre-Conditions: **(optional)**	A confirmed scoring event and current score snapshot have been received.
Post Conditions: **(optional)**	Player can distinguish base, 2x, and 3x results and never sees a false deduction below $0.
Other attributes: **(optional)**	- None

Requirement Title: **(*required)**	Level Won, Level Lost, and Game Won screens
Sequence No: **(*required)**	017
Short description: **(*required)**	Present outcomes and actions supplied by Win/Lose and GameController.
Detailed Description: **(*required)**	Graphics shall show separate result states: Level Won! with reached target/achieved amount and Next Level for Levels 1 and 2; Level Lost! with required/achieved amount, Time's Up! reason where applicable and Retry Level; and Game Won! at Level 3 with banked total and Play Again / Quit. Graphics only sends chosen navigation action to GameController; it never declares a win or starts the next level itself.
Pre-Conditions: **(optional)**	A typed WinLose result and allowed navigation actions exist.
Post Conditions: **(optional)**	Result is readable, navigation sends controller commands and no fishing continues behind the overlay.
Other attributes: **(optional)**	- None

Requirement Title: **(*required)**	Freeze scene and display timeout overlay
Sequence No: **(*required)**	018
Short description: **(*required)**	Make timer-expiry outcome visible while preventing background fishing.
Detailed Description: **(*required)**	Upon Game Controller Time Expired, Graphics shall stop rendering fish motion and cast/struggle progress, preserve a static frame of the current map, disable on-screen fishing inputs, and show a Time's Up! notification followed by Level Lost! and a Retry Level option. No AI movement, catch animation, or score mutation shall continue behind the overlay. The controller and AI handle stopping logic, Graphics reflect their terminal state.
Pre-Conditions: **(optional)**	Timer has expired and GameController closed the current attempt.
Post Conditions: **(optional)**	Scene is visibly frozen, controls unavailable, and level loss appears without ongoing fishing.
Other attributes: **(optional)**	- None

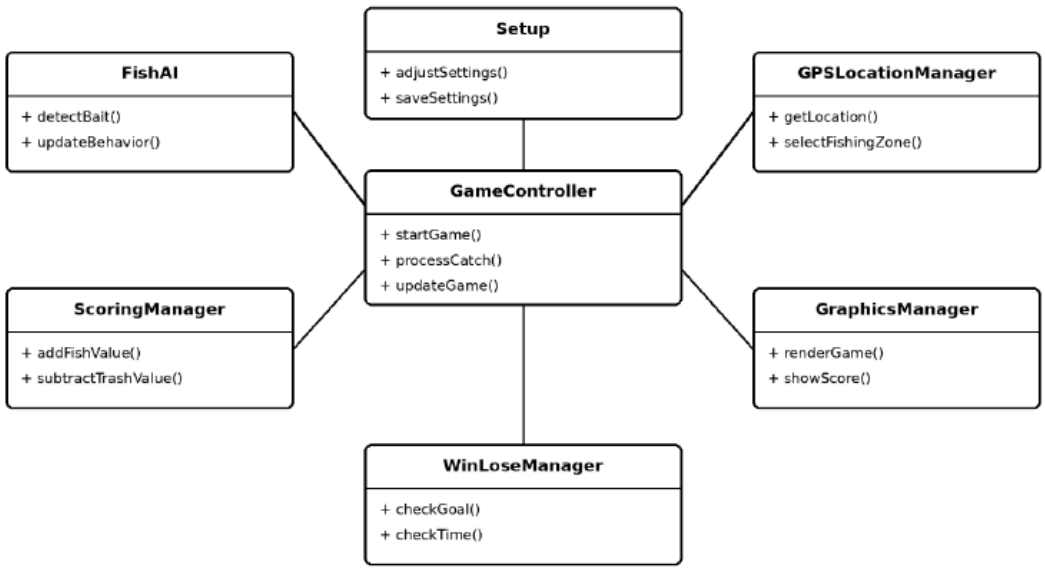
5. Software Processes and Infrastructure

5.1 Hardware and Infrastructure

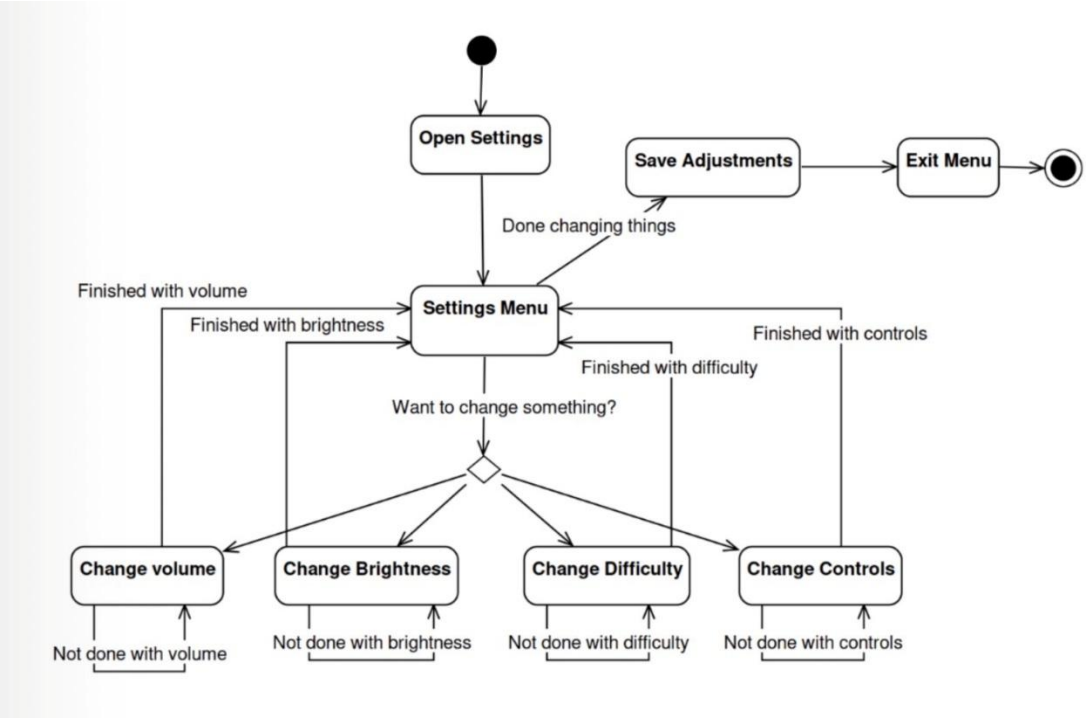
- **Platform:** Android mobile device or the Android emulator required by CSE 3310. Input is by touch only.
- **Android version / API level:** KitKat or higher version of android
- **Game engine:** Unity, provisional. The requirements are engine neutral.
- **Development hardware:** PCs or laptops capable of running Unity and the Android emulator.
- **Network:** The requirements describe no online features, accounts, or external services. The game runs entirely on the device.
- **Location services:** Not used. GPS, geolocation permissions, and GPSLocationManager are explicitly prohibited.
- **Audio:** Music and sound effects play through the game's own volume settings and do not change device system volume.

5.2 UML Diagrams

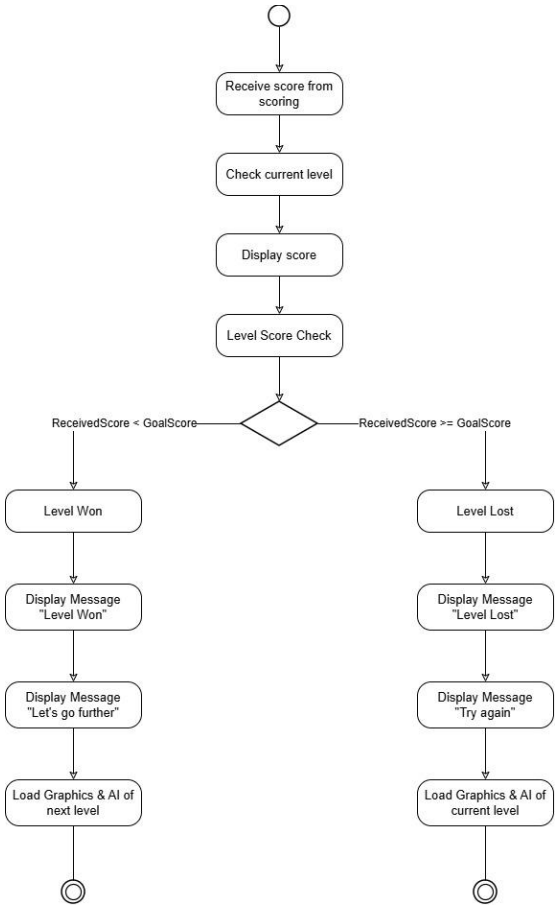
UML_CLASS_SYSTEM



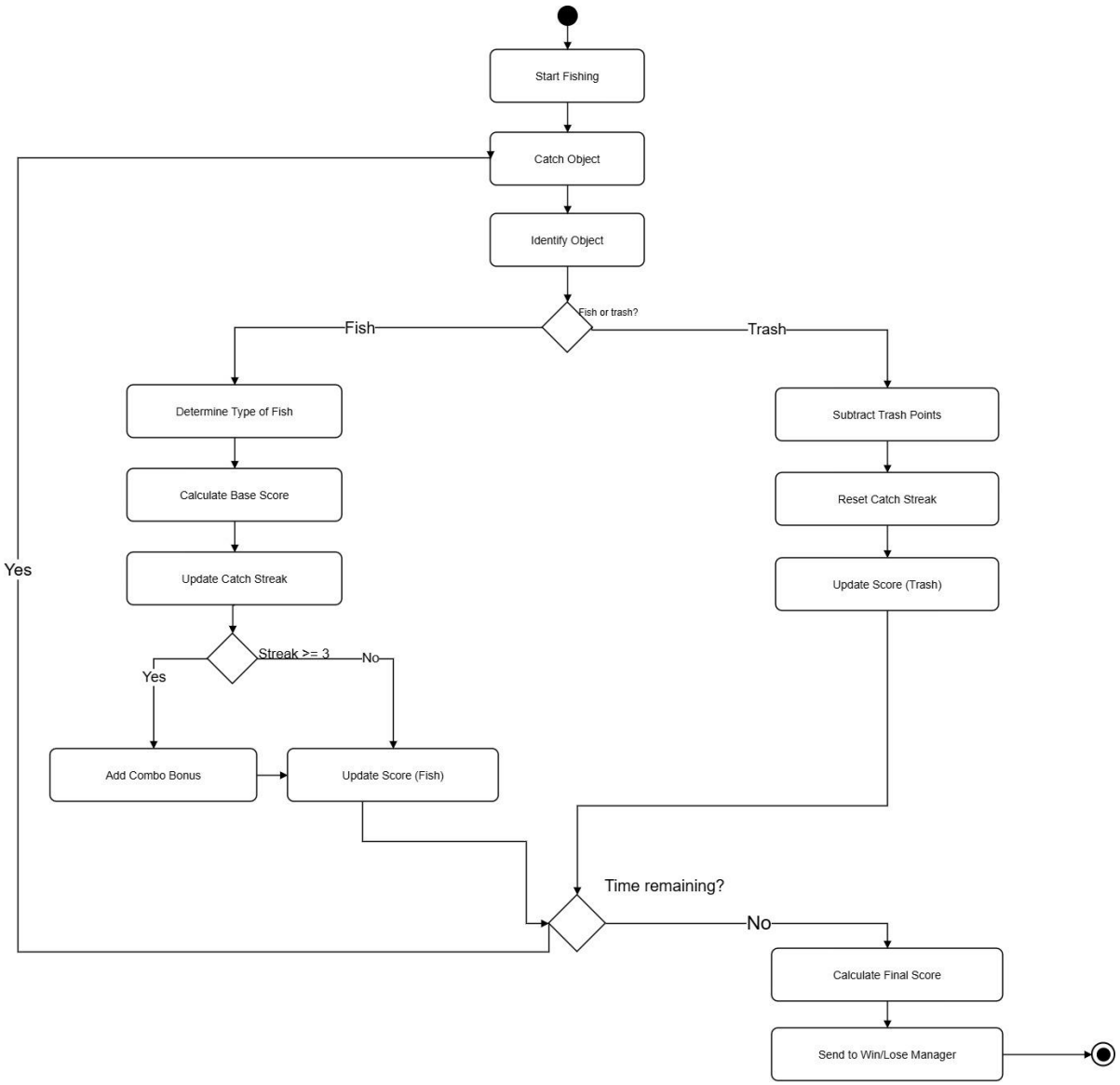
UML_ACTIVITY_SETUP



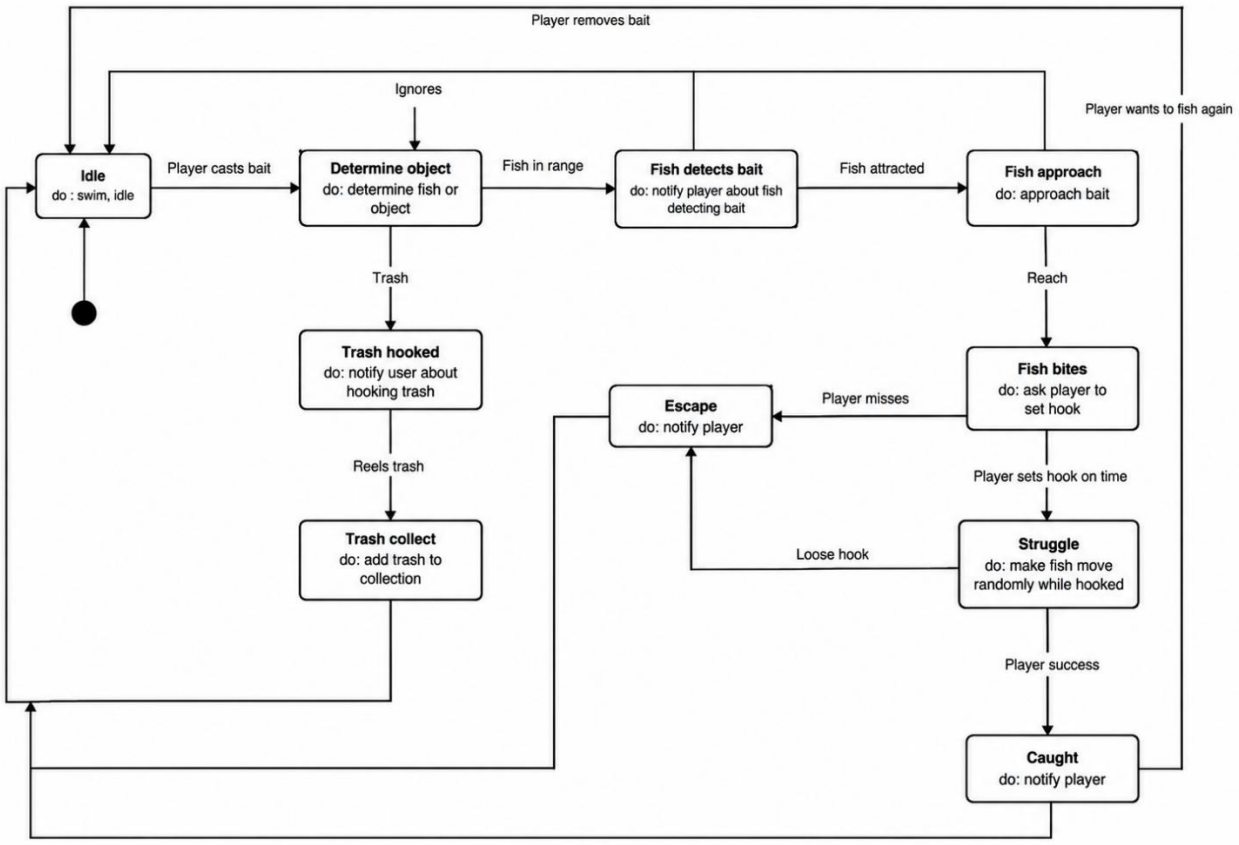
UML_STD_WIN/LOSE



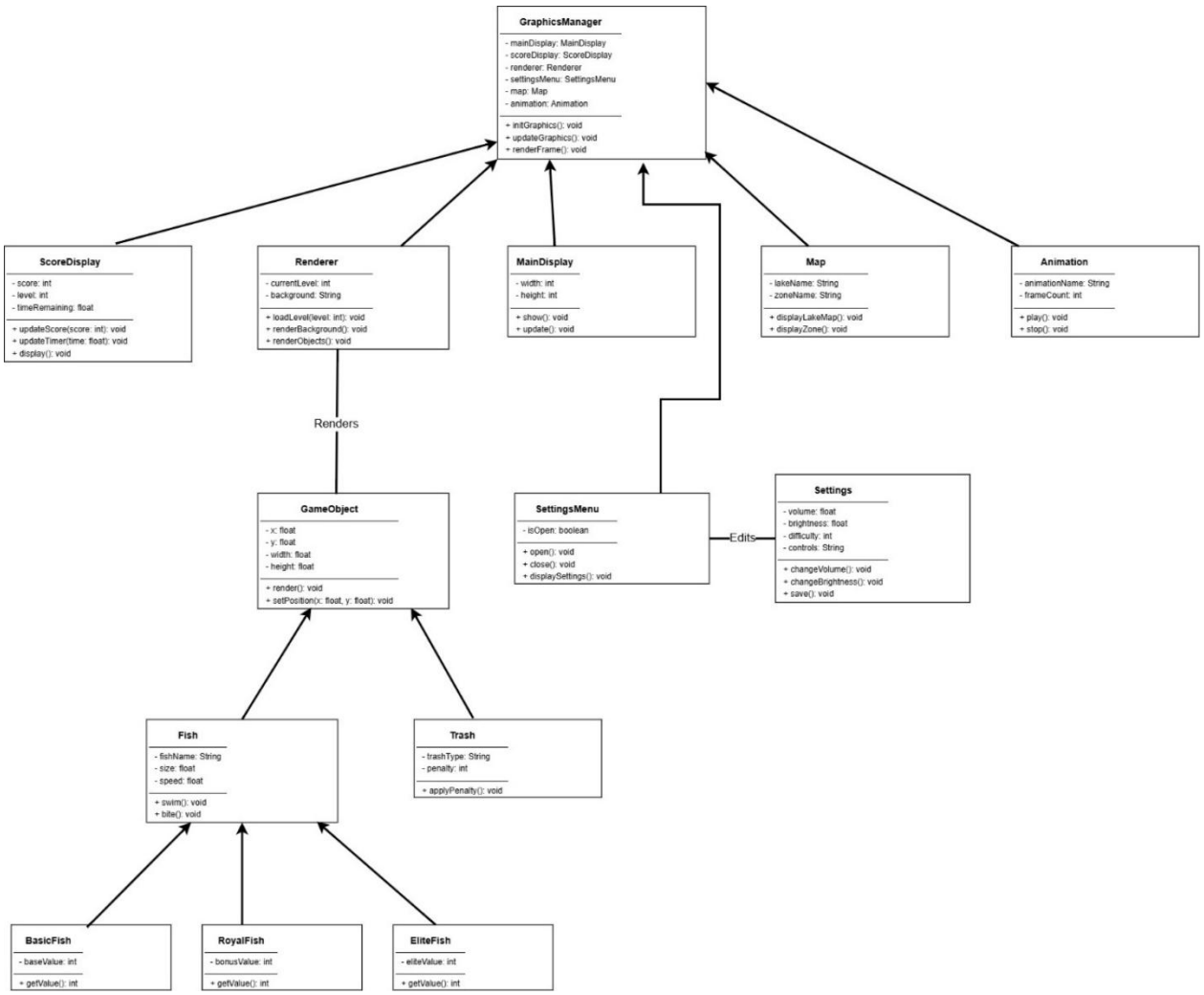
UML_ACTIVITY_SCORING



UML_STD_AI


Player removes bait
Ignores
Player wants to fish again
Idle
do : swim, idle
Player casts bait
Determine object
do: determine fish or object
Fish in range
Fish detects bait
do: notify player about fish detecting bait
Fish attracted
Fish approach
do: approach bait
Trash
Reach
Trash hooked
do: notify user about hooking trash
Fish bites
do: ask player to set hook
Escape
do: notify player
Player misses
Reels trash
Player sets hook on time
Trash collect
do: add trash to collection
Loose hook
Struggle
do: make fish move randomly while hooked
Player success
Caught
do: notify player

UML_CLASS_GRAPHICS



5.3 Conceptual Data Model

Use of Unity and Android Studio.

5.4 Screen Shots

Not available at this time.

5.5 Test Plan

A test plan will be provided at a later stage of the project.

6. Assumptions and Constraints

6.1 ASSUMPTIONS

The following is a list of assumptions:

- Each level uses a 600-second active-play timer. The duration is configurable. The timer pauses during Settings, loading, and result overlays.
- Escape probabilities are Basic 10%, Royal 20%, Elite 30%, decided once during the struggle.
- Fish rarity is 60% Basic, 30% Royal, 10% Elite.
- Difficulty matrix for movement, hook window, and spawn interval is Easy 0.75×/ 3 s/ 2 s, Medium 1.0×/ 2 s/ 3 s, Hard 1.25×/ 1 s/ 4 s. Medium is the default.
- A fish in bait range approaches with 75% probability and ignores with 25%. The detection radius is set during implementation.
- Only one fish or trash encounter owns the bait at a time.
- Each encounter is trash with 15% probability and fish with 85%.
- Controls are Cast, Hook, and hold-to-Reel buttons. A valid reel is 2 continuous seconds.
- A missed hook or escaped fish ends the current combo. Ignored bait and empty casts do not.
- Cleared-level money is kept for the current session only. A fresh launch starts at Level 1 with $0. Only Settings persist.
- Buttons are Next Level, Retry Level, Play Again, and Quit.
- At timeout, in-progress actions are canceled. Only events committed to Scoring strictly before the boundary count. A win check runs immediately after each committed score event.
- The three maps are fixed and affect visuals only. They do not change fish rarity.
- If a level overshoots its target, the actual amount is banked. There is no $50,000 cap.
- Defaults are music 100%, effects 100%, brightness 100%, sensitivity 50%, button size 100%, difficulty Medium. Sliders run 0–100%, and button size runs 75–150% in 5% steps. Brightness affects game visuals only.
- Quitting ends gameplay and does not bank an unfinished level’s earnings.
- Combo totals are cumulative. Only the difference is credited on each catch.
- The $100 trash penalty applies only to current-level earnings.
- Money is awarded only on AI-confirmed catches. Each event has a unique ID and is ignored if duplicated, stale, or late.
- No timer runs while a map loads between levels.
- Trash is a single collectible type with no inventory or disposal screen.
- Probabilities, timing, and targets have not been play tested. Reaching $25,000 in 10 minutes may be too hard.

6.2 CONSTRAINTS

The following is a list of constraints:

- The team lacks Android development skills.
- The system must run on the Android emulator required by CSE 3310.
- Controls are touch-only.
- No GPS, device location, or location permissions may be used.
- The game has exactly three levels and three fixed maps.

6.3 OUT-OF-SCOPE MATERIAL

The following is a list of “out of scope” material:

- A shop, fish-to-money conversion screen, or selling step
- Inventory and fishing-rod upgrades
- GPS, geolocation, or player-selected fishing zones
- A “Continue Fishing” prompt after each catch
- Story, narrative, or cutscenes
- Resume-after-closing for game progress
- Support for any platform other than Android
- Real-money transactions or in-app purchases
- Additional levels beyond three

7. Delivery and Schedule

Task/Milestone Description	Anticipated Start Date	Anticipated End Date	Status {Complete, In Progress, To Be Completed-TBC}	Comments (Deliverable plus the team member working on the listed item)
Prepare UML diagrams	9/10/2026	9/24/2026	Complete	Deliverable ➔ UML document (TEAM)
Create an official SRA document	9/27/2026	10/29/2026	Complete	Deliverable will be the SRA document. All stakeholders agree on the content of the SRA by signing in section 8.
Scoring	10/30/2026	11/27/26	TBC	Hiya Kavalanekar
Win/Lose	10/30/2026	11/27/26	TBC	Hiya Kavalanekar
Setup	10/30/2026	11/27/26	TBC	Alyssa Aguilar
AI	10/30/2026	11/27/26	TBC	Prasil Niraula
Graphics	10/30/2026	11/27/26	TBC	Jayneshkumar Patel
Test Plan Delivery	11/1/2026	11/24/2026	TBC	The Team Deliverable will be the Test plan document.
External Documentation (i.e., User Manual)	11/26/2026	12/1/2026	TBC	The team Deliverable will be a User Manual.
Final Milestone: Project Delivery and Team Presentation	11/26/2026	12/1/2026	TBC	The Team Deliverable will be the final project binder plus the product demo

8. Stakeholder Approval Form

Stakeholder Name	Stakeholder Role	Stakeholder Comments	Stakeholder Approval Signature and Date
Bahram Khalili	Client		
Fahmid Rifat	Client Project Manager		
Hiya Kavalanekar	Developer		
Alyssa Aguilar	Developer		
Prasil Niraula	Developer		
Jayneshkumar Patel	Developer		

Appendix:

None